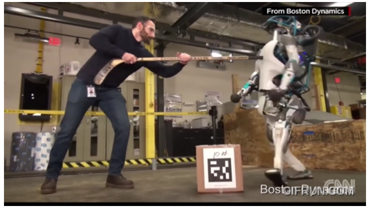

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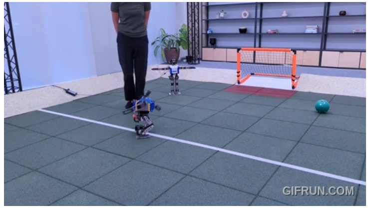

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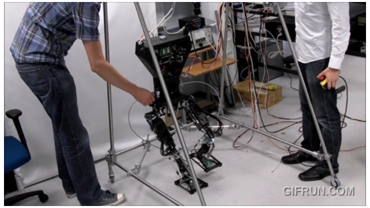

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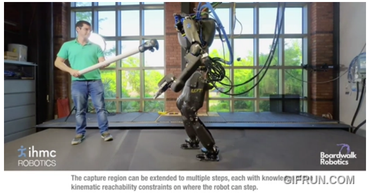

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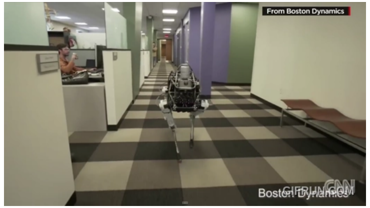

12

[1]CNN [2]OP3 Soccer [3]TU Delft [4]CNN [5]OSU [6]Unitree [7]OCP3 Soccer [8]CNN [9]Corridor Digital [10]IHMC Robotics [11]Agility Robotics [12]CNN

Project Neo

Dominic Zahn, 5/5/2026

Neo-Construct

Template

A ready-to-go robotics optimal control setup in the form of a Docker container. The Neo Construct encapsulates a whole ROS2 Gazebo environment to solve robotics problems with a...

Dockerfile ☆ 2

https://gitlab.kit.edu/kit/iar-hcr/frameworks/neoConstruct

Pinocchio
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RBDL

RViz

Pinocchio
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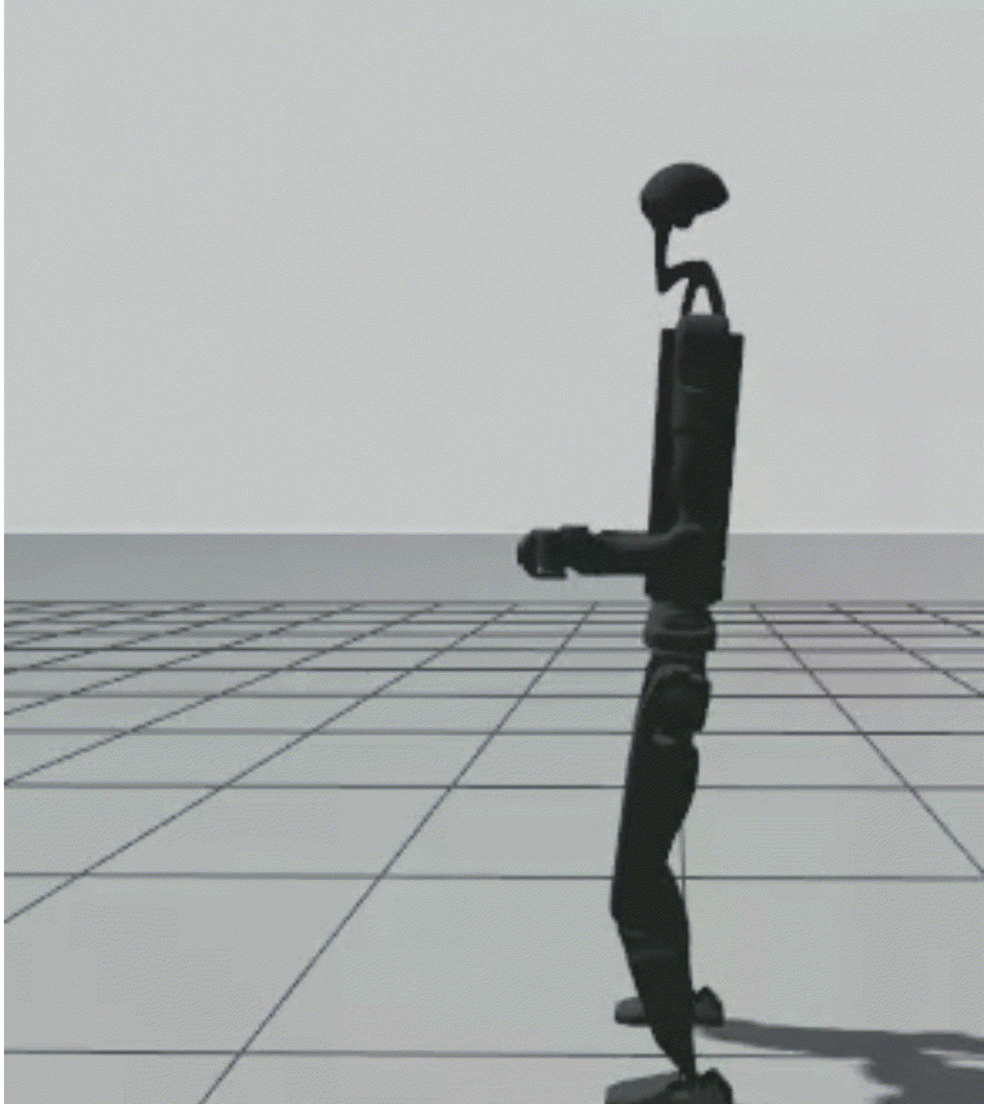
NLOPT

He1nz
the Chosen One



Questions?

OCP Formulation

$$x(t) = \begin{bmatrix} q(t) \\ \dot{q}(t) \end{bmatrix} \tag{1}$$

$$u(t) = \ddot{q}(t) \tag{2}$$

$$f_{expl}(x(t), u(t), t) = \begin{bmatrix} \dot{q}(t) \\ \ddot{q}(t) \end{bmatrix} \tag{3}$$

$$\min_{x(.),\, u(.)} \int_0^{T_f} \omega_s l_s(x(t), u(t)) + \omega_h l_h(x(t), t) + \omega_r l_r(u(t)) dt \quad \text{(Obj.)}$$

$$\begin{aligned}
\text{s.t.} \quad & \begin{bmatrix} q_0 \\ 0 \end{bmatrix} = x(0) & & \text{(Cons. 0)} \\
& \begin{bmatrix} \underline{q}(t) \\ -v_{max} \end{bmatrix} \leq x(t) \leq \begin{bmatrix} \overline{q}(t) \\ v_{max} \end{bmatrix} & \forall t \in [0, T_f] \quad & \text{(Cons. 1)} \\
& -a_{max} \leq u(t) \leq a_{max} & \forall t \in [0, T_f] \quad & \text{(Cons. 2)} \\
& p_{stab}(t) \in \mathrm{P} & \forall t \in [0, T_f] \quad & \text{(Cons. 3)}
\end{aligned}$$